

DS-GA 1018: Homework 4

Due Friday November 18th at 5:00 pm

Problem 1 (8 points): We have shown that the initialization of the EM algorithm limits will determine the final result. We have largely presented this as a limitation of EM, but in this problem we will show that, for discrete Hidden Markov Models, it can be a strength. Imagine that we have an initial guess for our dHMM parameters:

$$\boldsymbol{\theta}_0 = \{\boldsymbol{\pi}_0, \mathbf{A}_0\}. \quad (1)$$

We also have access to our observation probability distribution function $p(\mathbf{x}_t | \mathbf{z}_t)$. Our goal is to show that if $\mathbf{A}_{ij,0} = 0$ then $\mathbf{A}_{ij,1} = 0$. Each subproblem will work us through the steps.

i. (3 points): Assume that $\mathbf{A}_{ij,0} = 0$ and that the other values of $\mathbf{A}_{jk,0}$ have arbitrary values. Solve for $p(z_{t,i}, z_{t+1,j} | \mathbf{x}_{1:T})$.

$$\begin{aligned} p(z_t, z_{t+1} | \mathbf{x}_{1:T}) &= \frac{p(\mathbf{x}_{1:T} | z_t, z_{t+1}) p(z_t, z_{t+1})}{p(\mathbf{x}_{1:T})} \\ &= \frac{p(\mathbf{x}_{1:t} | z_t) p(\mathbf{x}_{t+1} | z_{t+1}) p(\mathbf{x}_{t+2:T} | z_{t+1}) p(z_{t+1} | z_t) p(z_t)}{p(\mathbf{x}_{1:T})}, \\ &= \frac{\alpha(z_t) p(\mathbf{x}_{t+1} | z_{t+1}) \beta(z_{t+1}) p(z_{t+1} | z_t)}{p(\mathbf{x}_{1:T})} \end{aligned}$$

thus we know:

$$p(z_t = i, z_{t+1} = j | \mathbf{x}_{1:T}) \propto \alpha(z_t) p(\mathbf{x}_{t+1} | z_{t+1} = j) \beta(z_{t+1} = j) p(z_{t+1} = j | z_t = i)$$

Thus $A_{ij,0} = 0$, we have $p(z_{t+1} = j | z_t = i) = 0$, thus $p(z_t = i, z_{t+1} = j | \mathbf{x}_{1:T}) = 0 \quad \forall t = 1, \dots, T$.

ii. (3 points): Assume that $\mathbf{A}_{ij,0} = 0$ and that the other values of $\mathbf{A}_{mn,0}$ for $m, n \neq i, j$ have arbitrary values. Solve for $\mathbf{A}_{ij,1}$.

$$A_{kk',n} = \frac{\sum_{t=1}^T p(z_{t,k'}, z_{t-1,k} | \mathbf{x}_{1:T}, \boldsymbol{\pi}_{n-1}, \mathbf{A}_{n-1})}{\sum_{j=1}^K \sum_{t=1}^T p(z_{t,j}, z_{t-1,k} | \mathbf{x}_{1:T}, \boldsymbol{\pi}_{n-1}, \mathbf{A}_{n-1})},$$
 and that we have proved in problem (i) that $p(z_t = i, z_{t+1} = j | \mathbf{x}_{1:T}) = 0 \quad \forall t = 1, \dots, T$, thus the numerator evaluates to 0 and $A_{ij,1} = 0$.

iii. (2 points): What can you say about the solution for $\mathbf{A}_{ij,n}$. When might this behavior be considered a benefit of the EM algorithm?

1. If it is known that certain transitions (e.g., $i \rightarrow j$) are physically or logically impossible, setting $A_{ij,0} = 0$ ensures that these transitions are never learned, reducing the parameter search space.
2. For applications where the true transition matrix is sparse (e.g., sequences with limited state transitions), initializing with $A_{ij,0} = 0$ for impossible transitions helps the EM algorithm converge faster and reduces overfitting.
3. By enforcing $A_{ij,n} = 0$ for specific transitions, the learned model remains consistent with prior knowledge, making the results easier to interpret.

Problem 2 (14 points): Consider a discrete HMM model with K unique classes for the latent state. Imagine that the transition matrix has the form:

$$A_{ij} = \frac{1}{K}. \quad (2)$$

Assume that the initial state vector π has arbitrary values and that we have access to our observation probability distribution function $p(\mathbf{x}_t | \mathbf{z}_t)$.

i. (4 points): For $t > 0$, simplify the recursive equation for $\hat{\alpha}(\mathbf{z}_t)$ as much as possible.

We start from the definition:

$$c_t \hat{\alpha}(\mathbf{z}_t) = p(\mathbf{x}_t | \mathbf{z}_t) \sum_{\mathbf{z}_{t-1}} \hat{\alpha}(\mathbf{z}_{t-1}) p(\mathbf{z}_t | \mathbf{z}_{t-1})$$

where we know that $p(\mathbf{z}_t | \mathbf{z}_{t-1}) = A_{ij} = \frac{1}{K}$, $\sum_{\mathbf{z}_{t-1}} \hat{\alpha}(\mathbf{z}_{t-1}) = c_{t-1}$ and $\sum_{\mathbf{z}_t} \hat{\alpha}(\mathbf{z}_t) = c_t$. Thus we could get the following:

$$\begin{aligned} c_t \hat{\alpha}(\mathbf{z}_t) &= p(\mathbf{x}_t | \mathbf{z}_t) \sum_{\mathbf{z}_{t-1}} \hat{\alpha}(\mathbf{z}_{t-1}) \frac{1}{K} \\ \hat{\alpha}(\mathbf{z}_t) &= p(\mathbf{x}_t | \mathbf{z}_t) \frac{c_{t-1}}{K \cdot c_t} \quad \forall t > 0 \end{aligned}$$

ii. (6 points): For $t < T$, simplify the recursive equation for $\hat{\beta}(\mathbf{z}_t)$ as much as possible. *Hint: Start from $t = T - 1$ and use that to derive your solution for the remaining t . Don't forget your solution to c_t .*

We start from

$$\begin{aligned} c_{t+1} \hat{\beta}(\mathbf{z}_t) &= \sum_{\mathbf{z}_{t+1}} \hat{\beta}(\mathbf{z}_{t+1}) p(\mathbf{x}_{t+1} | \mathbf{z}_{t+1}) p(\mathbf{z}_{t+1} | \mathbf{z}_t) \\ &= \sum_{\mathbf{z}_{t+1}} \hat{\beta}(\mathbf{z}_{t+1}) p(\mathbf{x}_{t+1} | \mathbf{z}_{t+1}) \frac{1}{K} \end{aligned}$$

Thus we will have :

$$\hat{\beta}(\mathbf{z}_t) = \sum_{\mathbf{z}_{t+1}} \hat{\beta}(\mathbf{z}_{t+1}) p(\mathbf{x}_{t+1} | \mathbf{z}_{t+1}) \frac{1}{K \cdot c_{t+1}} \quad \forall t \leq T - 1$$

iii. (2 points): Solve for $p(z_t | \mathbf{x}_{1:T})$.

$$\begin{aligned} p(z_t | \mathbf{x}_{1:T}) &= p(\mathbf{x}_t | z_t) \frac{c_t - 1}{K \cdot c_t} \cdot \left(\sum_{z_{t+1}} \hat{\beta}(z_{t+1}) p(x_{t+1} | z_{t+1}) \frac{1}{K \cdot c_{t+1}} \right) \\ &= p(\mathbf{x}_t | z_t) \frac{c_t - 1}{K^2 \cdot c_t \cdot c_{t+1}} \cdot \left(\sum_{z_{t+1}} \hat{\beta}(z_{t+1}) p(x_{t+1} | z_{t+1}) \right) \end{aligned}$$

iv. (2 points): What is the intuition behind this result? *Hint: the values of $\mathbf{A}$ should factor into your argument.*

1. Since $A_{ij} = \frac{1}{K}$, the transition probabilities are uniform, meaning no specific transitions are preferred. This causes the influence of transitions to be distributed equally among all states.
2. The posterior probability $p(z_t | \mathbf{x}_{1:T})$ is a combination of contributions from observations at t , the forward information $(x_1, \dots, x_t)$, and the backward information $(x_{t+1}, \dots, x_T)$. The scaling factors c_t ensure numerical stability.

Problem 3 (10 points): In lecture we showed how we can compose different Gaussian Process kernel to generate a new Gaussian process kernel. When do this composition, we are implicitly stating that, given two kernels κ_1 and κ_2 that:

$$\kappa_+(\mathbf{x}, \mathbf{x}') = \kappa_1(\mathbf{x}, \mathbf{x}') + \kappa_2(\mathbf{x}, \mathbf{x}') \quad (3)$$

$$\kappa_\times(\mathbf{x}, \mathbf{x}') = \kappa_1(\mathbf{x}, \mathbf{x}') \times \kappa_2(\mathbf{x}, \mathbf{x}'), \quad (4)$$

are both valid kernel functions.

i. (5 points): Prove that if κ_1 and κ_2 are valid kernel functions, then κ_+ is a valid kernel function.

Suppose the corresponding matrix for κ_1 and κ_2 , then they are positive semi-definite since both kernels are valid. Since we know summation of PSD matrices are still PSD, which means the matrix of $\kappa_1 + \kappa_2$ is PSD, which implies $\kappa_1 + \kappa_2$ is a valid kernel.

ii. (5 points): Prove that if κ_1 and κ_2 are valid kernel functions, then $\kappa_{\times}$ is a valid kernel function.

We assume that we are proving element-wise product of two valid kernels. The proof goes as follows: Suppose κ_1 corresponds to matrix A and κ_2 corresponds to matrix B and $\kappa_{\times}$ corresponds to matrix C . Now since A, B are PSD, we have that $A = D^{\top}D$ and $B = E^{\top}E$ where we have $A_{ij} = \sum_l d_{il}d_{jl}$ and $B_{ij} = \sum_k e_{ik}e_{jk}$. We want to show that matrix $C = A \odot B$ where $C_{ij} = A_{ij} \times B_{ij}$ has the following property: $\sum_{i,j} u_i C_{ij} u_j \leq 0$, and the proof goes as follows:

$$\vec{u}^{\top} C \vec{u} = \sum_{i,j} u_i C_{ij} u_j \quad (5)$$

$$= \sum_{i,j} u_i A_{ij} B_{ij} u_j \quad (6)$$

$$= \sum_{i,j} u_i \left(\sum_l d_{il} \cdot d_{jl} \right) \left(\sum_k e_{ik} \cdot e_{jk} \right) \quad (7)$$

$$= \sum_{k,l} \sum_{i,j} u_i d_{il} d_{jl} e_{ik} e_{jk} u_j \quad (8)$$

$$= \sum_{k,l} \left(\sum_i u_i d_{il} e_{ik} \right) \left(\sum_j u_j d_{jl} e_{jk} \right) \quad (9)$$

$$= \sum_{k,l} \left(\sum_i u_i d_{il} e_{ik} \right)^2 \quad (10)$$

$$\geq 0 \quad (11)$$

, which finishes the proof.